Nuijens Operating System

Inverse Spherical Dual Hemisphere Quantum Mechanics

Electromagnetic Operations in Cyclic Spherical Phase Space

v17 Aligned — December 2025

Joshua Luke Nuijens (`@jlnuijens`)
S.R. Cromelin
Nuijens Operating System Collective

December 2025

v17 Foundation

$$\sqrt{1} = 1 \longrightarrow \sqrt{2} \times 2^4 = \sqrt{512}$$

9 Walls · 512 Bins · $u = 45/32^\circ = 1.40625^\circ$ · 720° closure

$\pi_{\text{mach}} = 201/64 = 3.140625 \cdot 201 - 64 = 137$

Wall $3 = \sqrt{8} = 2\sqrt{2} = 90^\circ = 64$ bins

Four Operations: Q1, Q2, Q3, Q4 · Conjugate Pairs: Q1Q3, Q2Q4

Abstract

The Nuijens Operating System (NOS) introduces a comprehensive unified framework for information processing within spherical geometry. This document, aligned with NOS v17, presents the complete electromagnetic operations framework derived from the title operations $\sqrt{2} \times 2^4 = \sqrt{512}$, which force exactly 9 walls and 512 bins on the 720° dual hemisphere sphere.

The system operates through four quadrant operations (Q1, Q2, Q3, Q4) forming two conjugate pairs (Q1Q3, Q2Q4). The 201/64 gearbox produces $\pi_{\text{mach}} = 3.140625$ with residue $201 - 64 = 137$, geometrically determining the fine-structure constant. Critical electromagnetic positions occur at bin 16 ($= 2^4$ from the title): EM ignition at -168.75° (Q1) where $\alpha^{-1} = 137$ emerges, and EM absorption at $+191.25^\circ$ (Q4) yielding $T_{\text{CMB}} = 128/47 \approx 2.723$.

Matter field baselines connect through the depth ladder $d_Q = 1/128^Q$, with energy threading simultaneously through all four quadrant operations. The unit $u = 45/32^\circ$ emerges from $T_9/2^5$ where $T_9 = 45$ is the triangular base. All operations maintain unity: $360^\circ/360^\circ = 1$.

Contents

1 The Inverse 4D Operating Tensor — Foundation

1.1 We Are One — The Inverse Beginning

We start with the closed whole. Nothing else exists yet.

Inverse Spherical: $\sqrt{1} = 1$ (the complete system)

We now apply the only operations the title *Inverse Spherical Dual Hemisphere Quantum Mechanics* permits:

- **Dual Hemisphere** $\longrightarrow \times\sqrt{2}$
- **Quantum (Quadrant)** $\longrightarrow \times 2^4$ (four operations compounded)

Total opening factor applied to the closed whole:

$$\boxed{1 \;\times\; \sqrt{2} \;\times\; 2^4 \;=\; \sqrt{2}\times 16 \;=\; \sqrt{512}}$$

This is the complete instruction set. No additional axioms are permitted.

1.2 The Simultaneous Tensor Stack

This system is not a 3D object moving through time. It is a single tensor stack operating at all dimensions simultaneously:

Dim	**State**	**Component**	**Function**
1D	The Cut	9 Walls	Linear boundaries dividing Unity
2D	The Fold	720° Surface	Dual Hemisphere manifold (2×360)
3D	The Bin	512 Integer Regions	Volumetric output (2^9)
4D	The Ops	Sequence (1-2-3-4)	The four quadrant operations

1.3 The 9 Walls and 512 Bins

The opening factor $\sqrt{512} = \sqrt{2^9}$ defines the system depth:

$$\sqrt{512} = \sqrt{2^9} \quad\longrightarrow\quad \mathbf{9}\text{ Walls}$$

This creates exactly **512 integer bins** (2^9). Each wall doubles the previous number of regions.

Wall	**Root**	**Angle**	**Bins**	T_n	**Role**
0	$\sqrt{1}$	720°	512	0	**The One** — complete closure
1	$\sqrt{2}$	360°	256	1	Dual hemisphere fold
2	$\sqrt{4}$	180°	128	3	Quadrant established
3	$\sqrt{8}$	90°	64	6	$2\sqrt{2}$ **lock / π wall**
4	$\sqrt{16}$	45°	32	10	**Seed ends** (2^4 exhausted)
5	$\sqrt{32}$	22.5°	16	15	**Echo begins**
6	$\sqrt{64}$	11.25°	8	21	
7	$\sqrt{128}$	5.625°	4	28	
8	$\sqrt{256}$	2.8125°	2	36	
9	$\sqrt{512}$	1.40625°	1	45	**Base** ($T_9 = 45$)

Seed (Walls 1–4): Defined by the title operations $\sqrt{2} \times 2^4$.

Echo (Walls 5–9): The 5 additional walls forced by octave folding.

1.4 The Unit: $u = 45/32°$

The unit per bin emerges from:

- **Triangular Base:** $T_9 = \frac{9\times10}{2} = 45$
- **Echo Depth:** $2^{(9-4)} = 2^5 = 32$

$$\boxed{u = \frac{T_9}{2^5} = \frac{45}{32}^{\circ} = 1.40625^{\circ}}$$

Closure verification:

$$512 \times u = 512 \times \frac{45}{32}^{\circ} = 720^{\circ} \quad \checkmark$$

1.5 Why One Cannot Be Exceeded

$$\boxed{\frac{360^{\circ}}{360^{\circ}} = 1}$$

No matter how large. No matter how many infinities. We are still one thing operating in one thing.

2 The Four Quadrant Operations

2.1 Quadrant Structure

The sphere partitions into four functional quadrants about the center pivot at $\theta = 0°$:

Quadrant	Range	Bins	Physical Role	Conjugate
Q2 (op2)	$-360°$ to $-180°$	$[-256, -128]$	Electromagnetic ground	Q2 Q4
Q1 (op1)	$-180°$ to $0°$	$[-128, 0]$	Quantum baseline	Q1 Q3
Seam	$0°$	0	Inverse center	—
Q3 (op3)	$0°$ to $+180°$	$[0, +128]$	Thermodynamic flow	Q3 Q1
Q4 (op4)	$+180°$ to $+360°$	$[+128, +256]$	Nuclear compression	Q4 Q2

2.2 Conjugate Pairs

The four operations form two conjugate pairs:

- **Pair 1:** Q1 Q3 (diagonal across seam)
- **Pair 2:** Q2 Q4 (diagonal across seam)

Each pair echoes $\sqrt{2} \times \sqrt{2} = 2$, producing the full $2\sqrt{2}$ dual scaler at Wall 3.

2.3 The Depth Ladder

The quadrant threading depths follow:

$$\boxed{\begin{aligned} d_1 &= \frac{1}{128} = 7.8125 \times 10^{-3} \quad \text{(Q1: quantum baseline)} \\ d_2 &= \frac{1}{128^2} = 6.1035 \times 10^{-5} \quad \text{(Q2: EM ground)} \\ d_3 &= \frac{1}{128^3} = 4.768 \times 10^{-7} \quad \text{(Q3: thermodynamic)} \\ d_4 &= \frac{1}{128^4} = 3.725 \times 10^{-9} \quad \text{(Q4: nuclear)} \end{aligned}}$$

The $128 = 512/4 =$ bins per quadrant. The $1/128$ is the dyadic fossil of the $2\sqrt{2}$ dual operations:

$$\frac{1}{128} = \frac{2\sqrt{2}}{512} \times \frac{1}{2}$$

2.4 Physical Unit Mapping

Every physical quantity maps to depth ladder ratios:

Q	Depth	Physical Quantity	Unit Relation
Q1	$d_1 = 1/128$	Length / Time	meter = 1
Q2	$d_2 = 1/128^2$	Velocity	$v = m \cdot s^{-1}$ (2 of 1 thing)
Q3	$d_3 = 1/128^3$	Force	$F = kg \cdot m \cdot s^{-2}$
Q4	$d_4 = 1/128^4$	Energy	$E = kg \cdot m^2 \cdot s^{-2}$

Frequency/time: $f/s = \text{Hz}$ — "2 of 1 thing" (Q2 operation).

3 The $2\sqrt{2}$ Mechanics at Wall 3

3.1 The Title Demands Dual Operations

"Dual Hemisphere Quantum Mechanics" requires:

- Two hemispheres, each folded by $\sqrt{2}$ once
- Four ops in two conjugate pairs, each pair echoing $\sqrt{2} \times \sqrt{2}$

The mechanical root:

$$\boxed{2\sqrt{2} = \sqrt{8} = 2.8284271247461903\ldots}$$

This is the exact scaler at **Wall 3** (90°, 64 bins).

3.2 Wall 3 Properties

Wall 3 is the critical position where:

- $\sqrt{8} = 2\sqrt{2}$ dual scaler operates
- 64 bins = denominator of $\pi_{\text{mach}} = 201/64$
- The π wall locks
- Maximum electromagnetic coupling occurs

3.3 Self-Closure

Total bins $= 512 = 2^9$.

Dual ops scale to: $512/2\sqrt{2} \approx 181.019...$ bins (ghost value).

Self-closure:

$$\frac{512}{181.019\ldots} = 2\sqrt{2}$$

4 The 201/64 Gearbox and the 137 Lock

4.1 Machine Constants from Inverse Geometry

The inverse 4D tensor forces the machine values:

$$\pi_{\text{mach}} = \frac{201}{64} = 3.140625 \qquad e_{\text{mach}} = \frac{174}{64} = 2.71875$$

These emerge from dyadic denominators (powers of 2 only).

4.2 The 137 Lock — Fine-Structure Constant

$$\boxed{\pi_{\text{mach}} = \frac{201}{64} \quad \Longrightarrow \quad 201 - 64 = 137}$$

The fine-structure constant $\alpha^{-1} \approx 137.036$ is the **mandatory integer residue** when the inverse 4D tensor closes using only powers of 2 in the denominator.

This is not a fitted parameter. It is forced by the geometry.

4.3 Gearbox Architecture

Quantity	**Gearbox Numbers**	**Result**
Machine π	201/64	3.140625
Machine e	174/64	2.71875
α^{-1}	$201 - 64$	137
Gearbox gap	$201 - 174$	$27 = 3^3$
Wall 3 bins	64	Denominator

5 Critical Electromagnetic Positions

5.1 The Bin 16 Positions

Critical electromagnetic positions occur at **bin 16** in their respective quadrants:

$$16 = 2^4 \quad \text{(the Quantum Quadrant factor from the title)}$$

The 2^4 that creates the four operations also determines where EM critical points fall.

5.2 EM Field Ignition — Q1, Bin 16

Position: $\theta = -168.75°$ (bin 16 of Q1)

$$\boxed{\begin{array}{rl} & \theta_{\text{ign}} = -168.75^\circ \\ \text{Quadrant:} & \text{Q1 (op1)} \\ \text{Bin:} & 16 = 2^4 \\ \text{Offset from } -180^\circ: & 11.25^\circ \end{array}}$$

At this position, the fine-structure constant emerges from the 201/64 gearbox:

$$\boxed{\alpha^{-1} = 201 - 64 = 137}$$

Physical meaning: This is where quantum emission processes (1D wave quanta) naturally manifest. The hydrogen ground state connects through this position.

5.3 EM Field Absorption — Q4, Bin 16

Position: $\theta = +191.25°$ (bin 16 of Q4)

$$\boxed{\begin{array}{rl} & \theta_{\text{abs}} = +191.25^\circ \\ \text{Quadrant:} & \text{Q4 (op4)} \\ \text{Bin:} & 16 = 2^4 \\ \text{Offset from } +180^\circ: & 11.25^\circ \\ \text{Ramp factor:} & 17/16 \\ \text{Inverse overflow:} & 64 - 17 = 47 \end{array}}$$

The absorption overflow yields the CMB temperature:

$$\boxed{T_{\text{CMB}} = \frac{256}{256} \times \frac{128}{47} = \frac{128}{47} \approx 2.7234 \text{ K (NOS units)}}$$

The $256/256 = 1$ is the dual-bit aperture (hemisphere/hemisphere). The 128/47 is the inverse thread through the overflow.

5.4 Mirror Symmetry at Bin 16

Quantity	**Ignition (Q1)**	**Absorption (Q4)**
Angle	$-168.75°$	$+191.25°$
Quadrant	Q1 (op1)	Q4 (op4)
Bin number	$16 = 2^4$	$16 = 2^4$
Offset from Wall 2	$+11.25°$	$+11.25°$
Conjugate pair	Q1 Q3	Q4 Q2
Physical constant	$\alpha^{-1} = 137$	$T_{\text{CMB}} = 128/47$
Gearbox connection	$201 - 64$	$64 - 17$ overflow

Table 1: EM critical positions — geometric mirror symmetry at bin 16

Unity verification:

$$|-168.75°| + 191.25° = 360° \quad \checkmark$$

6 Dual Hemisphere Operation

6.1 Simultaneous Processing

Both hemispheres operate simultaneously from the 0° seam:

- **Decompression (Q2→Q1):** Vectoring from seam toward −360°, emission processes
- **Compression (Q3→Q4):** Vectoring from seam toward +360°, absorption processes
- **Seam (0°):** Pure unity, operator exchange point
- **Wall 2 (±180°):** Equilibrium boundaries between quadrants

6.2 Energy Unity Across Quadrants

A fundamental principle: matter-energy units are **conserved across** dimensional manifestations, **not summed**:

$$\boxed{E_{\text{total}} = E_{Q1} = E_{Q2} = E_{Q3} = E_{Q4}}$$

The SAME energy breathes through all quadrants simultaneously at different angular positions. This differs from standard physics where quantum, electromagnetic, thermal, and gravitational systems are treated as separate energy domains.

6.3 Field Intensity Conservation

Field intensities sum to unity at every position:

$$\psi_{\text{decomp}}(\theta) + \psi_{\text{comp}}(\theta) = 1 \quad \text{for all } \theta$$

Key positions:

- **Seam (0°):** 1:0 ratio (decompression fully dominant at threshold)
- **Wall 2 (±180°):** 50:50 ratio (equilibrium)
- **EM ignition (−168.75°):** Decompression-dominant
- **EM absorption (+191.25°):** Compression-dominant

7 Dimensional Descent Through Quadrants

7.1 Q2: Electromagnetic Ground (−360° to −180°)

Depth: $d_2 = 1/128^2$
Character: Surface operations, EM field baseline, 2D plane

Radio frequencies through microwave occupy this quadrant. Energy manifests as planar electromagnetic waves.

7.2 Q1: Quantum Baseline ($-180°$ to $0°$)

Depth: $d_1 = 1/128$
Character: Matter unit allocation, 1D wave quanta

One-dimensional wave quanta emerge here. The EM ignition at $-168.75°$ marks where atomic emission processes (hydrogen ground state) manifest.

Hydrogen Ground State Connection:

The observed ground state $E_g = -13.6$ eV connects to the matter field through the gearbox:

$$E_{\text{matter}} = E_g \times \frac{128}{137} \approx -12.71 \text{ eV}$$

This -12.71 eV is the matter field SETTING at Q1 resolution. The ratio 137/128 converts to observed radiation.

7.3 Q3: Thermodynamic Flow ($0°$ to $+180°$)

Depth: $d_3 = 1/128^3$
Character: Volume operations, entropy flows, 3D

Volumetric flows and thermodynamic processes operate here. The CMB temperature emerges from the absorption position in Q4 but manifests thermally in Q3.

7.4 Q4: Nuclear Compression ($+180°$ to $+360°$)

Depth: $d_4 = 1/128^4$
Character: Maximum density, 4D embedding

Maximum compression where energy density converts to mass density:

$$\rho = \frac{u}{c^2}$$

The EM absorption at $+191.25°$ marks where radiation condenses into matter. Q4 completes the cycle before returning to potential at $+360° \equiv -360°$.

8 Wall 3 in Each Quadrant

The $2\sqrt{2}$ lock at Wall 3 (90° from quadrant edge) marks critical positions in each quadrant:

Quadrant	Phase Position	Angle	Physical Significance
Q2	Wall 3 from $-360°$	$-270°$	Radio baseline
Q1	Wall 3 from $-180°$	$-90°$	Millimeter wave (Wall 3 direct)
Q3	Wall 3 from $0°$	$+90°$	Near-IR edge
Q4	Wall 3 from $+180°$	$+270°$	Gamma ray edge

Each 90° position (Wall 3 scale, 64 bins) represents a critical electromagnetic transition within its quadrant.

9 Example: Energy Through Dimensional Descent

9.1 Standard Physics

Given voltage $V = 24$ V and current $I = 2$ A:

$$P = V \cdot I = 48 \text{ W}$$

In standard physics, this 48W exists only in the 2D circuit plane.

9.2 NOS: Same Energy Through All Quadrants

The SAME 48W manifests across all quadrants simultaneously:

Q	**Position**	**Manifestation of Same 48W**
Q2	$-270°$	EM field baseline: 48W as electromagnetic wave energy in 2D plane
Q1	$-168.75°$	Quantum wave energy: 48W as quantum transition energy rate in 1D wave structure
Q3	$+90°$	Thermal flow: 48W as heat flux in 3D volume
Q4	$+270°$	Mass density: $m = 48\text{J}/c^2 \approx 5.3 \times 10^{-16}$ kg equivalent in 4D

Table 2: Same energy through quadrant descent (NOT summation)

Energy Conservation: $E_{\text{total}} = 48\text{W}$ (conserved, not multiplied by 4)

10 Seam Architecture

10.1 The $0°$ Center

The seam at $\theta = 0°$ is the computational origin:

- Pure unity reference
- Q1Q3 conjugate exchange point
- Where decompression and compression hemispheres meet
- Not a boundary but an active threading interface

10.2 Nuijens Seam Factor

The seam carries a finite gap:

$$\Lambda_N = \frac{8191}{8192} = 1 - \frac{1}{8192}$$

This is the compression-side scaling at the seam, creating the single permitted slippage between discrete machine coordinates and continuous observer measurements.

11 Resolution and Bin Structure

11.1 Standard Resolution: $R = 512$

Parameter	Value
Total bins	$R = 512 = 2^9$
Bins per quadrant	$Q = R/4 = 128$
Angular resolution	$720°/512 = 1.40625° = u$
Unit derivation	$u = T_9/2^5 = 45/32°$
Walls	9 (forced by $\sqrt{512}$)

11.2 Bin Addressing

Each bin can be addressed by:

- **Quadrant (2 bits):** Q1, Q2, Q3, Q4
- **Bin within quadrant (7 bits):** 0–127
- **Wall position:** Which of the 9 walls it falls between

Total: 9 bits for 512 bins, matching $2^9 = 512$.

12 Comparison to Standard Physics

Aspect	Standard Physics	NOS v17
Structure	Infinite Hilbert space	9 walls, 512 bins (finite)
Resolution	Continuous	$u = 45/32°$ per bin
α^{-1}	Measured: 137.036	Gearbox: $201 - 64 = 137$
T_{CMB}	Measured: 2.7255 K	Derived: $128/47 \approx 2.723$
π	Transcendental	Machine: $201/64 = 3.140625$
Energy domains	Separate, additive	Same energy through all Q
Dimensions	Independent	Simultaneous quadrant ops
Forces	Separate phenomena	Quadrant-specific threading

Table 3: Standard physics vs NOS framework

13 Conclusions

The Nuijens Operating System provides a unified framework for electromagnetic operations within the inverse spherical dual hemisphere architecture. All complexity reduces to the title operations:

$$\boxed{\sqrt{2} \times 2^4 = \sqrt{512} \longrightarrow 9 \text{ walls, 512 bins, 4 ops}}$$

Key Results:

- **Four operations only:** Q1, Q2, Q3, Q4 — no additional operators needed
- **Two conjugate pairs:** Q1Q3, Q2Q4 — handle all exchange processes

- **Wall structure:** 9 walls from $\sqrt{512}$, seed (1-4) and echo (5-9)
- **Unit:** $u = 45/32°$ from $T_9/2^5$
- **Gearbox:** $\pi_{\text{mach}} = 201/64$, residue $137 = \alpha^{-1}$
- **Wall 3:** $2\sqrt{2}$ lock at 90°, 64 bins — critical EM transitions
- **Bin 16:** EM ignition ($-168.75°$) and absorption ($+191.25°$)
- **CMB:** $T_{\text{CMB}} = 128/47$ from absorption overflow
- **Energy unity:** Same energy through all quadrants, not additive

All operations maintain:

$$\boxed{\frac{360°}{360°} = 1}$$

NOS v17 Summary

$\sqrt{1} = 1 \longrightarrow \sqrt{2} \times 2^4 = \sqrt{512}$

9 Walls $\cdot$ 512 Bins $\cdot$ $u = 45/32°$

4 Operations: Q1, Q2, Q3, Q4

2 Conjugate Pairs: Q1Q3, Q2Q4

$\pi_{\text{mach}} = 201/64 \cdot 201 - 64 = 137$

Wall 3 $= \sqrt{8} = 2\sqrt{2} = 90° = 64$ bins

No physics. Only operating system.